Episode 4: *The Rubber Plantation (□□□□□□□□□□)*

Full Game Design & Narrative Specification Document

1. Overview & General Info

- **Title: Boon Sop:** The Unbroken Debt — Episode 4: The Rubber Plantation
- **Time Period**: 1932 (French Protectorate Era in Cambodia)
- **Protagonist**: Sora (□□□□□□ - Age 24) – Armed with her sketchbook, oil lantern, magnetic compass, and the French colonial rubber stamp recovered from Episode 3.
- **Location**: Compagnie des Caoutchoucs Plantation Estate & Processing Cellars, Kampong Cham Province, Cambodia
- **Time of Event**: Dense Midnight Fog (11:30 PM – 3:30 AM)
- **Estimated Playtime**: 3 to 4 Hours
- **Genre**: 3D First-Person Atmospheric Survival Horror (Navigation, Mimic Evasion, & Environmental Puzzles)
- **Engine Target**: Babylon.js / PlayCanvas (WebGPU/WebGL native in-browser)
- **Visual Aesthetic**: Volumetric dense grey fog, muted olive greens, rusty iron oranges, decayed yellow French stucco, dirt brown trails, and warm amber lantern light spheres.

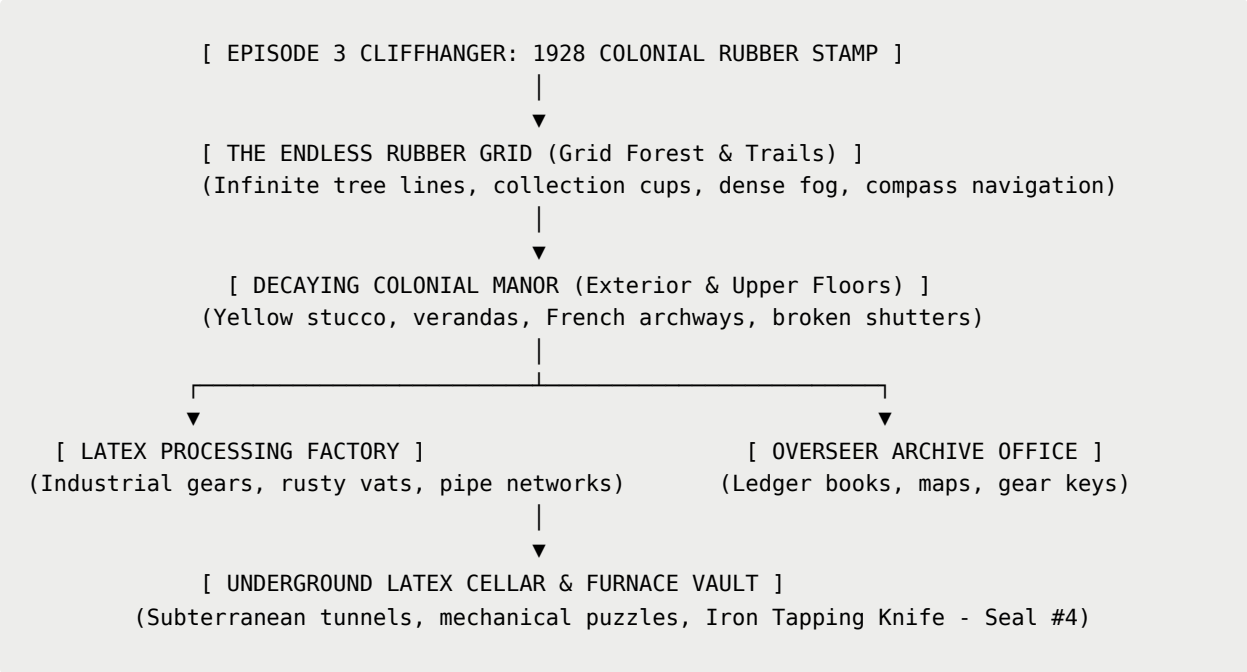
2. Character Profiles

- **Sora (□□□□□□□)**: Uncovering the deep historical roots of her family's curse, Sora uses her compass, map drawing skills, and acute hearing to navigate the disorienting, endless grids of rubber trees.
- **Overseer Jean-Luc (In Spirit / Notes)**: A long-deceased French plantation manager from the 1920s whose archived ledgers and journals detail the forced

labor conditions and the original contract signed by Sora's great-grandfather.

- **The Khmaoch Prey (□□□□□□□□□ - Forest Mimics)**: Tall, slender, disfigured woodland spirits inhabiting the dense rubber tree lines. They mimic familiar human voices—calling out Sora’s name using the voices of her grandmother, Uncle Sok, or late relatives—to lure her off safe paths into the fog.
- **The Neak Ta Shadow (□□□□□□)**: Manifesting as an inescapable dark presence weaving through the fog, attempting to crush Sora's lantern and break her internal sense of direction.

3. Environment & Spatial Layout



1. The Endless Rubber Grid (Chamkar Kaosou)

- **Structure**: Hundreds of tall, perfectly straight rubber trees planted in rigid, geometric grid rows. Each tree features diagonal spiral trunk cuts and a small, rusted clay cup attached to collect latex.
- **Lighting/Fog**: Extremely thick, volumetric ground-level fog limiting visibility to 3–4 meters. The player's lantern creates a soft sphere of amber light that illuminates floating fog motes.

2. Colonial Manor House (Phteah Barang)

- **Architecture**: A two-story French-Colonial mansion with peeling yellow stucco walls, arched verandas, wrought-iron balconies, and collapsed tiled roofs overgrown with jungle vines.
- **Interior Props**: Dust-covered gramophones, broken ceiling fans, overturned rattan armchairs, antique safes, and faded French maps of Kampong Cham.

3. Processing Factory & Underground Cellar

- **Structure**: Large industrial brick and corrugated iron processing sheds connected to a sprawling network of concrete underground cellars.
- **Props**: Rusted latex collection vats, iron pressure valves, heavy gear assemblies, overhead conveyor rails, and an industrial iron coal furnace.

4. Act Breakdown & Detailed Narrative

Act I: The Grid of Voices (0:00 – 1:00)

Narrative

Sora arrives at the edge of the vast Kampong Cham rubber estate late at night. A dense, unnatural fog has descended over the region, smothering all moonlight. Equipped with a brass magnetic compass and her oil lantern, she must venture into the disorienting grid of rubber trees to reach the abandoned colonial manor house.

As she moves through the silence of the trees, soft whispering voices begin echoing through the fog. She hears her Grandmother's voice calling out: *"Sora... over here... I'm lost in the trees..."* followed by Uncle Sok's voice calling from another direction.

Sora recalls a vital cultural taboo taught by her elders: If you hear a voice calling your name in the dark forest at night, never answer, and never turn around.

Gameplay Mechanics & Tutorial

- **Compass & Map Navigation**: Player uses a physical brass compass in hand alongside Sora's drawn grid map to navigate between numbered tree rows without getting lost in the fog loop.

- **The Audio Taboo ("Don't Look Back")**: When a *Khmaoch Prey* speaks in a familiar voice behind Sora, a distinct audio cue plays. The player must keep walking forward without turning the camera around or making sound until the voice fades.
- **Hand-Crank Lantern Fuel**: Sora's lantern uses oil that burns faster in dense fog. She must collect small oil cans attached to tapper stations throughout the forest.

Act II: Secrets of the Manor (1:00 – 2:30)

Narrative

Escaping the rubber grid, Sora reaches the overgrown French colonial manor house. Inside, she searches Overseer Jean-Luc's archived letters and estate ledgers from 1910.

She discovers the dark, tragic truth of her family line: During a devastating famine and cholera outbreak in 1910, her great-grandfather was a labor supervisor on this very estate. To save his dying family and secure wealth, he signed a blood contract with the ancient land guardian (Neak Ta), pledging the spiritual devotion and freedom of his bloodline for four generations. Uncle Sok had come here in 1928 to attempt to burn the original contract ledger, but was driven out by the entities guarding the cellar.

To access the underground cellar vault where the original contract tool is kept, Sora must solve the Industrial Gear Puzzle inside the estate's factory shed.

Gameplay Mechanics & Puzzles

- **Factory Pipe & Gear Puzzle**: Sora must inspect French mechanical schematics in her sketchbook, collect three missing iron gears from the manor rooms, and align pressure valves on the factory boiler to unlock the heavy cellar blast doors.
- **Interior Evasion**: Navigating dark, creaking manor corridors while avoiding shadow manifestations lurking inside window frames and wardrobe closets.

Act III: The Furnace Vault & Seal #4 (2:30 – 3:30)

Narrative

Sora descends into the subterranean latex cellars beneath the factory. The air is heavy with the smell of old sulfur and dried latex. In the deepest chamber, resting inside an open coal furnace frame, she finds the fourth contract artifact: an Iron Tapping Knife (Kabit Chrok Kaosou) encrusted with dried blood and engraved with binding runes.

As Sora picks up the knife, the central Neak *Ta* entity manifests in full physical force—blocking the main cellar exit with towering shadow tendrils while causing the industrial steam pipes to burst.

Gameplay Mechanics & Puzzles

- **The Smelting Counter-Ritual**: Sora must reignite the industrial coal furnace by loading coal fuel and using her matches. She must place the Iron Tapping Knife into the crucible and crank the bellows to melt the metal down completely.
- **High-Tension Pressure Defense**: While waiting for the knife to melt, Sora must manage steam valves to keep shadow entities at bay while maintaining lantern light under extreme spatial audio cues.
- **Escape Through Ventilation**: As the knife melts, Seal #4 breaks with a massive supernatural shockwave that collapses the main cellar stairwell. Sora must crawl through a collapsing concrete ventilation tunnel to reach the surface.

Episode Resolution & Cliffhanger

Sora crawls out of the ventilation shaft into the damp grass as the morning fog begins to lift at 4:00 AM. In her hand, she holds the molten silver-iron ingot from the furnace.

As she opens her sketchbook to review her notes, the molten ingot reacts with the paper: the bloody lines on her map fuse together, pointing far north to the sacred mountain of Phnom Kulen—where an ancient sunken temple holds the central altar of the Neak Ta, setting the stage for the final climax in **Episode 5: The Sunken Sanctuary**.

5. Audio & Visual Specification Sheet

Feature	Design Specification
Color Palette	Volumetric grey fog, muted olive green, rust orange, yellowed stucco, earthy brown, amber flame light.
Post-Processing	Volumetric fog shaders, depth-of-field blur, motion vignetting, dynamic lantern light attenuation.
Light Sources	Hand-cranked oil lantern, factory furnace glow, rusted wall braziers, pale fog-filtered moonlight.
Primary Ambience	Dead forest silence, dripping latex into clay cups, wind howling through rubber grids, creaking factory iron.
Horror Sound Cues	Voice mimicry (relatives calling Sora's name), steam pipe hisses, metal gear grinds, collapsing concrete.
UI Design	100% Diegetic. Compass held in hand; lantern oil level visible on mechanical glass tube; sketchbook satchel for map tracking.